

Practical 4

```
In [10]: import numpy as np
import random
from collections import defaultdict

GRID_SIZE = 4
ACTIONS = ['U', 'D', 'L', 'R']
TERMINAL_STATES = [(0, 0), (3, 3)]
GAMMA = 1.0
EPISODES = 5000

V = defaultdict(float)
returns_count = defaultdict(int)

# Policy
def policy(state):
    return random.choice(ACTIONS)

# Environment
def step(state, action):
    i, j = state

    if state in TERMINAL_STATES:
        return state, 0, True

    if action == 'U':
        next_state = (max(i-1, 0), j)
    elif action == 'D':
        next_state = (min(i+1, GRID_SIZE-1), j)
    elif action == 'L':
        next_state = (i, max(j-1, 0))
    else: # 'R'
        next_state = (i, min(j+1, GRID_SIZE-1))

    reward = -1
    done = next_state in TERMINAL_STATES
    return next_state, reward, done

# Monte Carlo Prediction
for episode in range(EPISODES):

    state = random.choice([
        (i, j) for i in range(GRID_SIZE)
        for j in range(GRID_SIZE)
        if (i, j) not in TERMINAL_STATES
    ])

    episode_data = []

    # Generate episode
    while True:
        action = policy(state)
        next_state, reward, done = step(state, action)
        episode_data.append((state, reward))
        state = next_state

        if done:
            break #

    # First-Visit MC
    visited_states = set()
    G = 0

    for t in reversed(range(len(episode_data))):
        state_t, reward_t = episode_data[t]
        G = reward_t + GAMMA * G
```

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        if state_t not in visited_states:
            visited_states.add(state_t)
            returns_count[state_t] += 1
            V[state_t] += (G - V[state_t]) / returns_count[state_t]

# ♦ Show Learning progress every 1000 episodes
if (episode + 1) % 1000 == 0:
    print(f"\nEpisode {episode+1}")
    for i in range(GRID_SIZE):
        row = []
        for j in range(GRID_SIZE):
            row.append(f"{V[(i,j)]:6.2f}")
        print(" ".join(row))

print("\nMonte Carlo Prediction completed.")

```

Episode 1000

```

    0.00  -8.08 -10.45 -12.37
    -7.21  -8.34  -9.48 -10.30
   -10.28  -9.07  -9.05  -8.33
   -12.19 -10.58  -7.48   0.00

```

Episode 2000

```

    0.00  -7.60 -10.51 -12.61
    -7.32  -8.21  -9.29 -10.22
   -10.31  -9.12  -8.66  -7.81
   -12.88 -10.73  -8.02   0.00

```

Episode 3000

```

    0.00  -7.60 -10.28 -12.26
    -7.15  -8.29  -9.07 -10.03
   -10.28  -9.05  -8.34  -7.44
   -12.66 -10.52  -7.66   0.00

```

Episode 4000

```

    0.00  -7.48 -10.09 -12.17
    -7.09  -8.19  -8.99 -10.09
   -10.18  -9.11  -8.43  -7.76
   -12.50 -10.41  -7.60   0.00

```

Episode 5000

```

    0.00  -7.52 -10.05 -12.20
    -7.01  -8.31  -9.07 -10.25
   -10.09  -8.99  -8.49  -7.79
   -12.21 -10.29  -7.61   0.00

```

Monte Carlo Prediction completed.